

GEOGRAPHY

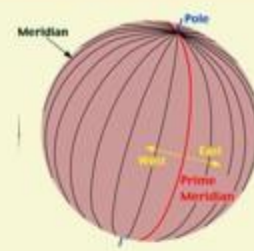
PHYSICAL GEOGRAPHY



MODULE - 11

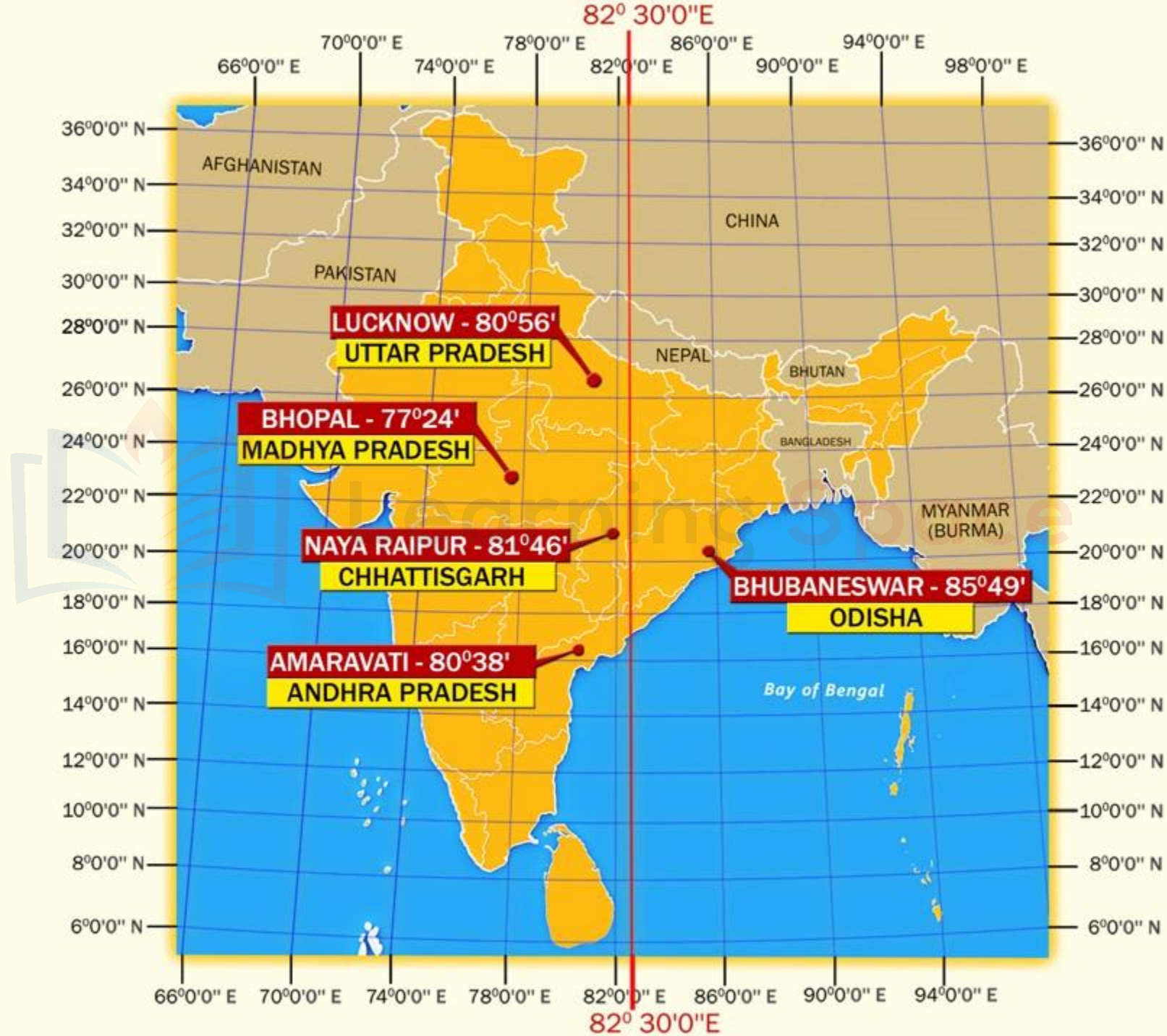
- LONGITUDES AND PRIME MERIDIAN
- INTERNATIONAL DATE LINE
- GMT, UTC AND LEAP SECOND
- VARIOUS TIME ZONES

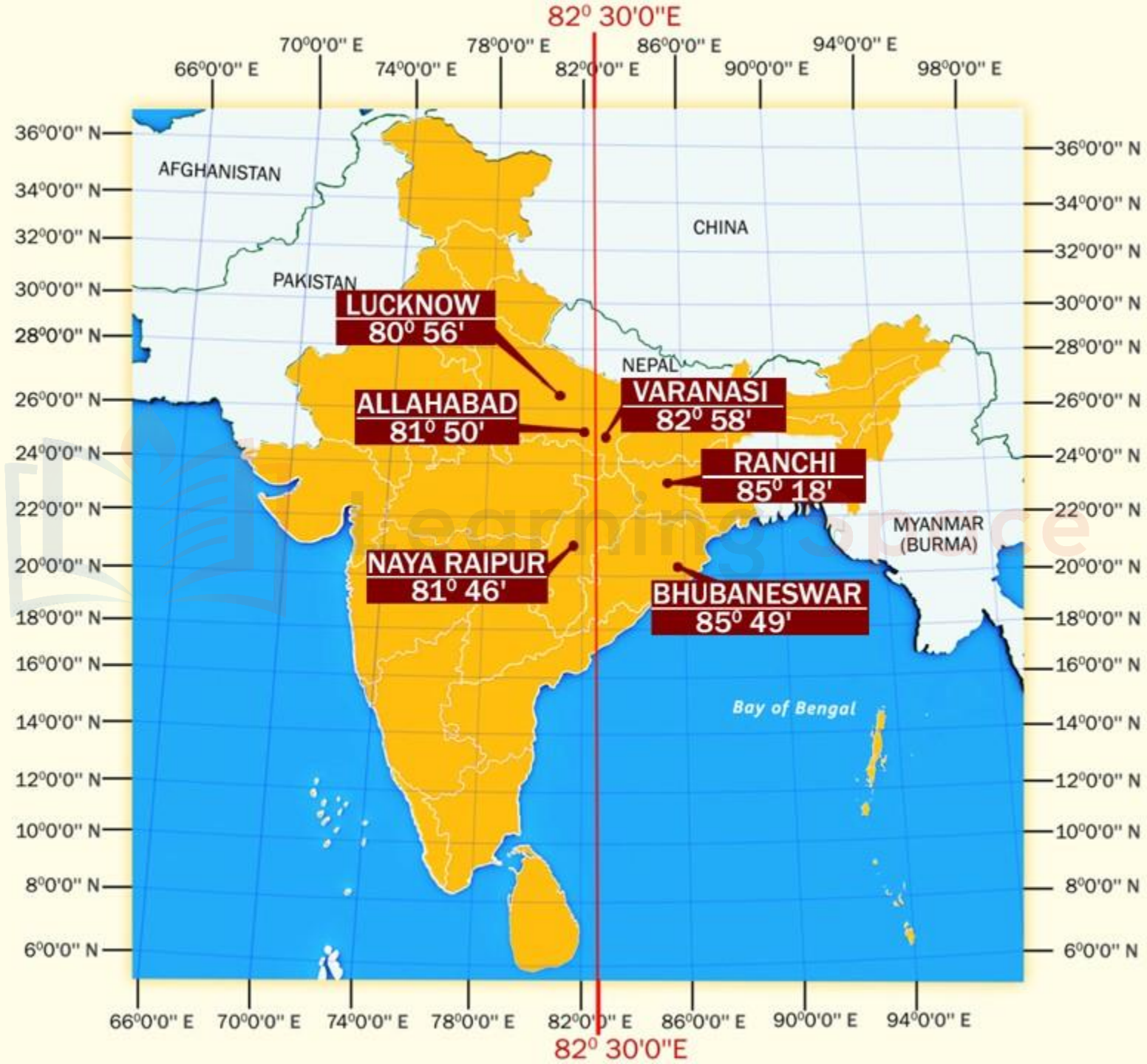
LONGITUDES



- 1** It is an angular distance measured in degrees along the equator, east or west of the Prime Meridian.
- 2** On the globe, longitude runs from pole to pole passing through the equator. These are called meridians.
- 3** In 1884, it was decided to choose zero meridian, the one which passes through the Royal Astronomical Observatory at Greenwich near London.
- 4** From the Prime Meridian, all other meridians radiate eastward and westward upto 180° .
- 5** The linear distance comes to be around 69 miles at equator for each degree.
- 6** At the poles, the linear distance is zero.









PRIME MERIDIAN

CONTINENTS & COUNTRIES

1. Europe

- United Kingdom
- France
- Spain

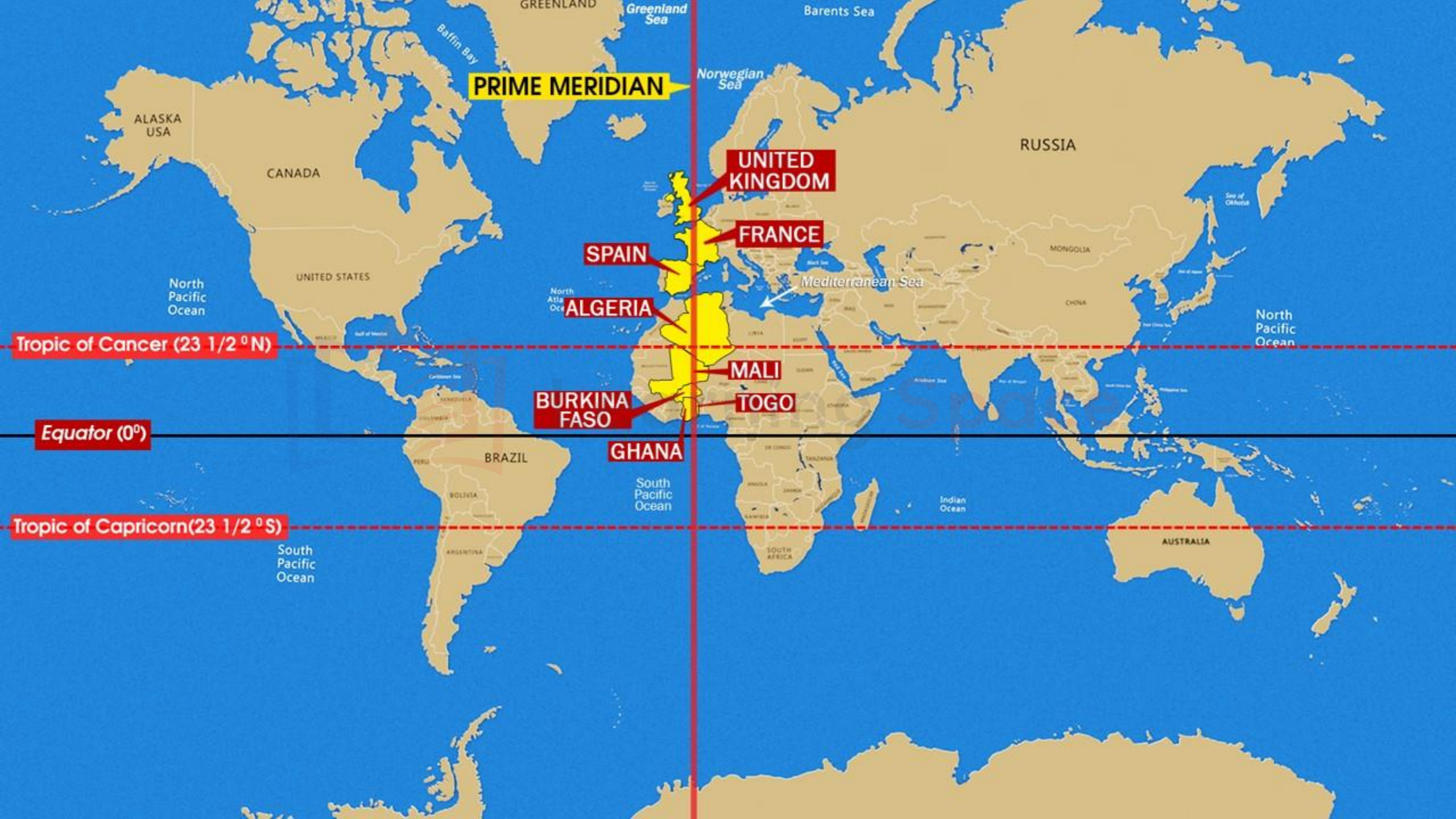
2. Africa

- Algeria
- Mali
- Burkina Faso
- Togo
- Ghana

3. Antarctica

- Antarctica

It also passes through water bodies such as Arctic Ocean, Norwegian Sea, Greenland Sea, Mediterranean Sea, Atlantic Ocean, Southern Ocean



PRIME MERIDIAN

UNITED KINGDOM

FRANCE

SPAIN

ALGERIA

MALI

TOGO

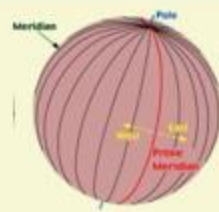
BURKINA FASO

GHANA

Tropic of Cancer (23 1/2 °N)

Equator (0°)

Tropic of Capricorn (23 1/2 °S)



LONGITUDE AND TIME



Russia extends through almost 165° of longitude and is divided into 11 time zones.

1 As the Earth moves from west to east, for every 15° if we go eastwards, local time is advanced by one hour. Conversely, if we go westwards, local time is retarded by one hour.

2 So the places east of Greenwich see the Sun earlier and gain time, where as the places west of Greenwich see the Sun later and loses time.

India accepted meridian of $82\frac{1}{2}^\circ$ E for the standard time, which is 5 hrs 30 min ahead of Greenwich Mean Time.

UT1



UTC



CO-ORDINATED UNIVERSAL TIME (UTC)

- It is a close variant of the Greenwich Mean Time or GMT at 0 degree longitude.
- UTC is defined by the International Telecommunications Union Recommendation, based on the International Atomic Time (TAI).
- According to the UTC, leap seconds are added to days at regular intervals to compensate for Earth's slowing rotation.



It is headquartered at Geneva, Switzerland

UT1



UTC



ASTRONOMICAL TIME AND UTC

- The Astronomical Time known as Universal Time (UT1) refers to the Earth's rotation around its own axis and determines the length of a day.
- However, the Earth's rotation around its own axis is not regular, as sometimes it speeds up and sometimes it slows down, due to various factors including the moon's gravitational Earth-braking forces that often results in ocean tides.
- As a result, Astronomical Time (UT1) gradually falls out of sync with Atomic time (UTC), and as and when the difference between UTC and UT1 approaches 0.9 seconds, a "Leap Second" is added to UTC through Atomic clocks worldwide.

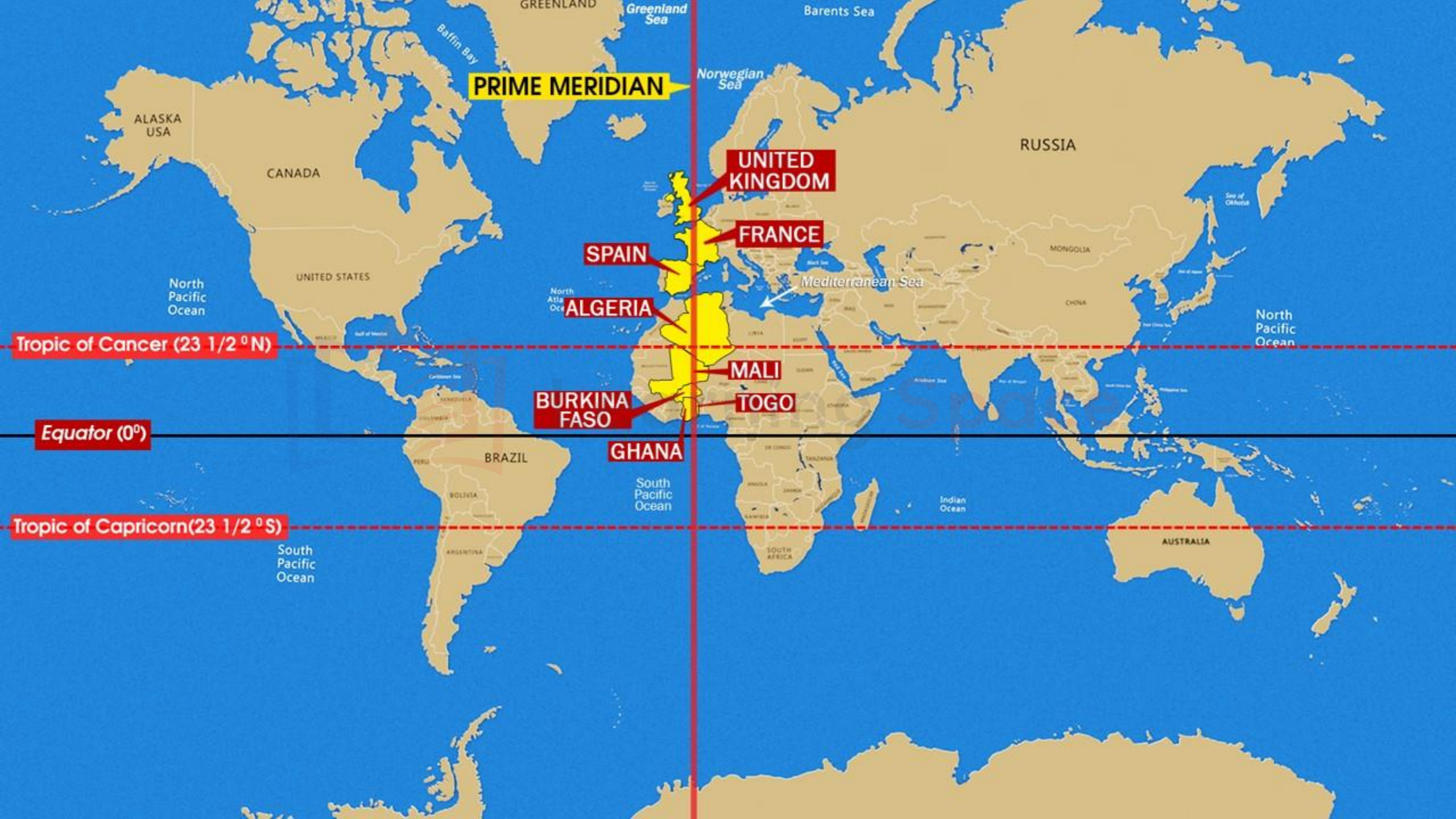
WHY THE LEAP SECOND IS ADDED?

- A “Leap Second” is added to Coordinated Universal Time (UTC) in order to synchronize clocks worldwide with the Earth's ever slowing rotation.
- UTC consists of a time scale that combines the output of more than 300 highly precise Atomic clocks worldwide.
- Atomic clocks are very accurate and are stable within 1 second over a period of millions of years.



UTC





PRIME MERIDIAN

UNITED KINGDOM

FRANCE

SPAIN

ALGERIA

MALI

TOGO

BURKINA FASO

GHANA

Tropic of Cancer (23 1/2° N)

Equator (0°)

Tropic of Capricorn (23 1/2° S)

DIFFERENCE BETWEEN GMT AND UTC



- GMT stands for Greenwich Mean Time and UTC for Coordinated Universal Time.
- Greenwich Mean Time is based on astronomical observations.
- Coordinated Universal Time (UTC) is the legal basis for time. It is a method of measuring time using atomic clocks.
- The time difference between UTC and GMT lies in fractions of a second.
- Therefore, for common purposes, both the times are considered the same. But, the time difference is important for scientific purposes.
- UTC is the time standard used for several Internet and World Wide Web standards.
- It can be said that the UTC is Internet based time standard whereas GMT is country based time standard.

GMT is a time zone and UTC is a time standard.

TOP FIVE COUNTRIES WITH MOST NUMBER OF TIME ZONES

1 FRANCE

- France has 12 time zones ranging from UTC-10 to UTC+12. This unusual span is due to France's scattered national territories.
- The main part of the country, France has only one time zone.
- The remaining 11 time zones are added by France's dependencies.
- The areas in French Polynesia in the Pacific Ocean are partly responsible for this.

2 RUSSIA

- The only country in the world that has all its time zones over the mainland is Russia.
- The biggest country in the world, Russia has 11 time zones.
- Russia is the country to have most consecutive number of time zones.
- 10 of the country's 11 time zones cover a contiguous landmass.
- Only the Russian exclave of Kaliningrad, wedged between Lithuania and Poland, breaks that pattern.



North
Pacific
Ocean

UNITED STATES

CHINA

AUSTRALIA

FRENCH POLYNESIA

SOUTH
PACIFIC
OCEAN



Norwegian Sea

Gulf of Bothnia

FINLAND

NORWAY

SWEDEN

Baltic Sea

KALININGRAD

RUSSIA

North Atlantic Ocean

North Sea

IRELAND

UNITED KINGDOM

NETHERLANDS

BELGIUM

GERMANY

LITHUANIA

POLAND

BELARUS

UKRAINE

FRANCE

SWITZERLAND

AUSTRIA

HUNGARY

CZECH REP

SLOVAKIA

SLOVENIA

CROATIA

SERBIA

ROMANIA

BULGARIA

ITALY

Adriatic Sea

Black Sea

GEORGIA

TOP FIVE COUNTRIES WITH MOST NUMBER OF TIME ZONES

3

USA

- United States of America has 11 time zones including its dependencies.
- The time zones range from UTC-12 to UTC+12. Two of these time zones (UTC-12 and UTC+12) for its dependencies are yet to become official for the country.
- The contiguous US has 4 standard time zones.

4

UK

- The mainland of the United Kingdom is not big at all.
- It is a small island to the northeast of the Atlantic Ocean.
- Due to their naval skills, the British had spread colonies all around the world.
- Thus, the country is still left with various time zones that are scattered across the planet.
- From the Pitcairn Islands in Pacific Ocean to the British Indian Ocean Territory, the Island colonisers have 9 time zones to follow.

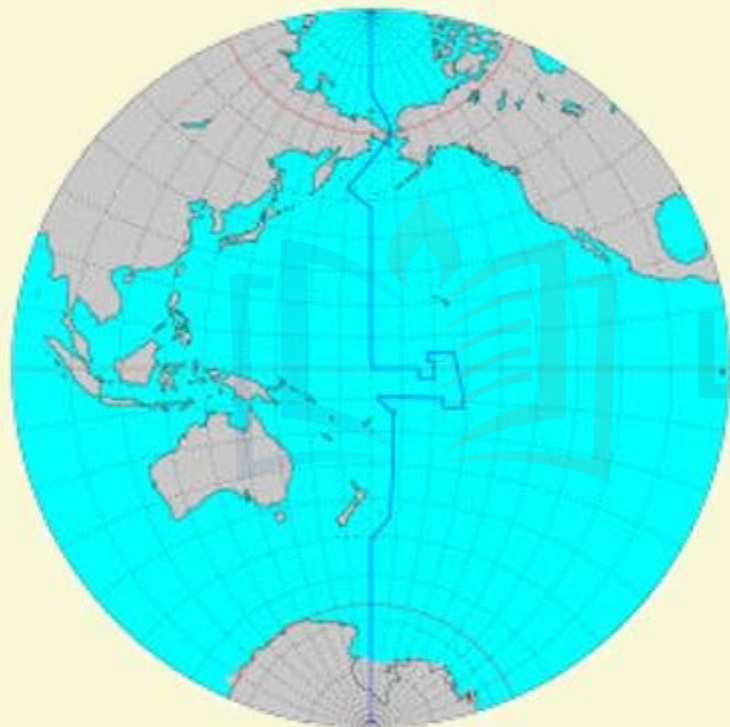
TOP FIVE COUNTRIES WITH MOST NUMBER OF TIME ZONES



5 AUSTRALIA

- Australia is the world's sixth largest country and has 9 time zones.
- There are 6 time zones in Australia. Adding the country's 3 dependencies brings the total to 9 time zones.
- The southern continent of Oceania is largely defined by Australia's land mass and the islands it takes under national territory.

INTERNATIONAL DATE LINE



- 1** International date line is the line, where the date changes by exactly one day, when it is crossed.
- 2** A traveller crossing the date line from east to west, loses a day and while crossing the date line from west to east, he gains a day.
- 3** That means, towards the Asian side, if the day is Sunday by crossing the line eastwards, it will be Saturday.

TIME AT VARIOUS PLACES WHEN IT IS 12 NOON IN LONDON







INTERNATIONAL DATE LINE



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